

Digibyte: Promoting Awareness and Habit Change for Sustainable Digital File Decluttering

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Abstract

This project addresses the environmental impact of unused digital files within universities. It is projected that just data centres alone, where cloud storage is managed, will consume the same amount of electricity as Japan by 2026. User research with university affiliates, comprising a survey of 34 and interviews with 5 participants, revealed a lack of awareness regarding the environmental costs, difficulties in managing files, and personal motivations rather than environmental for decluttering. In response, Digibyte was developed to centralise digital storage, while promoting awareness and habit change through game-like features and positive reinforcement. Digibyte empowers users by supporting their decluttering process and reflect on their digital storage habits. In the future, the platform can be expanded to other sectors to support sustainable digital file habits.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

behavioural change, digital waste, sustainability, universities

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1 Introduction

More data is being created and stored, impacting the environment. In 2020 alone, nearly 40 trillion gigabytes of data were created,

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equivalent to 1.7 MB per person per second [7]. Yet, the Global Databerg Report indicated that 52% of stored data is redundant, obsolete, or trivial [19]. While seemingly harmless, unused data contribute to digital waste. This project defines digital waste as any files that are no longer needed or used, such as duplicates and outdated documents. Storing these unused files, whether in the cloud or on physical devices, harms the environment due to the energy consumption of data centres and resource-intensive manufacturing of hard drives.

Data centres, where cloud storage is managed, consume vast amounts of energy and resources. The International Energy Agency (IEA) reported that in 2022 alone, data centres consumed an estimated 460 terawatt-hours (TWh) of electricity [1]. IEA projected that this consumption would grow to 1,000 TWh by 2026, which roughly equates to the electricity consumption of Japan [1]. Furthermore, vast amounts of clean water are used to efficiently cool the components in data centres [13]. For example, Apple's data centre in Mesa, Arizona, is straining water resources in a region already facing severe drought, with the year 2021 being the driest in 126 years [3, 14]. Even unused, digital waste takes up space in cloud storage, contributing to data centre operation and the consumption of energy and resources.

Beyond data centres' resource consumption, manufacturing hard drives for data centres and personal devices further harms the environment through carbon emissions. The manufacturing of a single gigabyte of flash memory contributes to 0.16 kg of CO₂ [17]. Moreover, Solid State Drives have a higher rate of infancy failure, which could lead to frequent replacements within data centres [12]. This frequent replacement creates a demand for more production and manufacturing of hard drives. The manufacturing harm extends to personal devices as well. A breakdown of Apple's emissions revealed an increase in total emissions across iPhone generations, which was largely attributed to manufacturing, especially due to upgrades such as storage capacity [4]. The accumulation of unused files on devices further fuels the demand for larger storage hardware, perpetuating a cycle of increased production and resource use.

1.1 University Context

University affiliates unconsciously store vast amounts of digital waste in physical and cloud storage, leading to carbon emissions and resource deprivation in communities. Generous cloud storage allowances, such as the 100 TB offered by Google Education and

Microsoft Education plans, can lead to the unmindful accumulation of digital files [10, 16]. Students accumulate a substantial number of digital files from the electronic resources provided by universities. Similarly, faculty and staff handle a diverse range of digital materials, including administrative documents and research files. As the stored data grows, so do the associated energy consumption and carbon emissions. To address this problem, this project sought to answer the question: “How can we support university affiliates in decluttering their digital waste while raising awareness of its environmental impact and encouraging long-term sustainable habits?” Research into behaviour change for environmental action has demonstrated that building awareness can encourage responsible habits [5]. This project’s design solution is Digibyte, a platform that centralises storage and supports decluttering while building awareness and empowering users to adopt long-term change. Digibyte aligns with the United Nations Sustainable Development Goal (SDG) #12: Responsible Consumption and Production, contributing to a more sustainable future by reducing the load on data centres and demand for larger storage devices, which harms the environment.

2 Related Work

Despite universities’ efforts to promote general sustainability, there is a lack of research and awareness regarding the environmental impact of excessive digital file storage caused by these institutions. UK universities aim to increase climate literacy and equip students and staff with the knowledge to understand their behaviours within the context of sustainability [18], yet the specific issue of digital file storage remains unexamined. Even with limited research, there are solutions like Clean My Mac and CCleaner that help users identify and remove system junk, temporary files, and unneeded applications [2, 9]. Similarly, cloud providers, like Google, offer tools to assist users in deleting files within their accounts, yet these solutions are limited to managing one account at a time [15]. All these solutions support users in lowering their storage usage within either physical or cloud storage. However, a gap remains in providing students and staff with a solution that centralises decluttering across all physical and cloud storage. Furthermore, the long-term solution to excessive digital files is not simply decluttering; it is fostering awareness and developing environmentally conscious behaviours to break the cycle of accumulation that burdens digital infrastructure and the communities in which they are built.

3 User Research

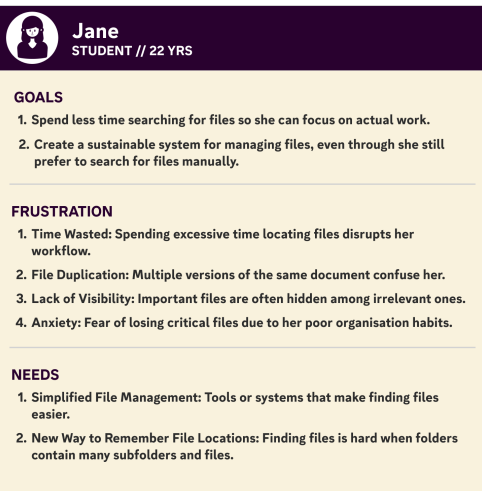
After identifying the problem, user research utilising surveys and semi-structured interviews was conducted to better understand the problem space and explore users’ current habits, pain points, and knowledge regarding the environmental impact of their digital waste.

3.1 Survey

A survey was conducted to better understand our target users’ file management habits and their awareness of the environmental impact. 34 anonymous responses were collected through convenience sampling.

It was found that the participants lacked awareness on the environmental impact of digital waste. Most of the participants were unaware of the energy consumption of unused data (84%) and the global carbon dioxide emission contributed by cloud storage (71%). One participant mentioned that answering the questionnaire made them contemplate about de-cluttering priority, suggesting the mere awareness of environmental impact can motivate users to reconsider their habits.

The questionnaire also revealed that participants struggled to locate files. More than half of the participants reported that they had struggled to find files in physical devices (52%) and cloud storage (66%). Notably, for cloud storage, 28% experienced such struggle on a daily and weekly basis. Participants expressed intense negative emotions, such as ‘despair’ and ‘get depressed’, when being asked what they would do when files cannot be located.



Jane
STUDENT // 22 YRS

GOALS

1. Spend less time searching for files so she can focus on actual work.
2. Create a sustainable system for managing files, even though she still prefer to search for files manually.

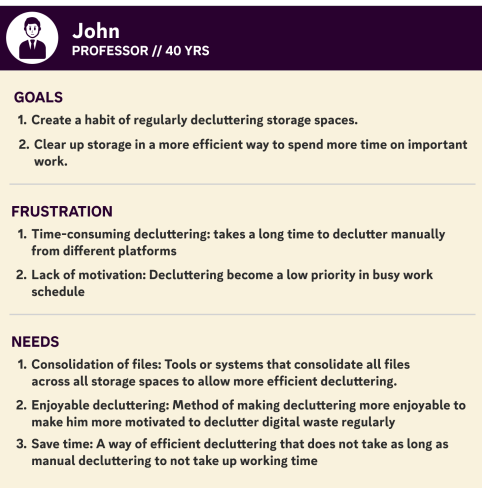
FRUSTRATION

1. Time Wasted: Spending excessive time locating files disrupts her workflow.
2. File Duplication: Multiple versions of the same document confuse her.
3. Lack of Visibility: Important files are often hidden among irrelevant ones.
4. Anxiety: Fear of losing critical files due to her poor organisation habits.

NEEDS

1. Simplified File Management: Tools or systems that make finding files easier.
2. New Way to Remember File Locations: Finding files is hard when folders contain many subfolders and files.

(a) User persona of a student overwhelmed by messy files.



John
PROFESSOR // 40 YRS

GOALS

1. Create a habit of regularly decluttering storage spaces.
2. Clear up storage in a more efficient way to spend more time on important work.

FRUSTRATION

1. Time-consuming decluttering: takes a long time to declutter manually from different platforms
2. Lack of motivation: Decluttering become a low priority in busy work schedule

NEEDS

1. Consolidation of files: Tools or systems that consolidate all files across all storage spaces to allow more efficient decluttering.
2. Enjoyable decluttering: Method of making decluttering more enjoyable to make him more motivated to declutter digital waste regularly
3. Save time: A way of efficient decluttering that does not take as long as manual decluttering to not take up working time

(b) User persona of a professor who finds cleaning up too time-consuming.

Figure 1: User Personas.

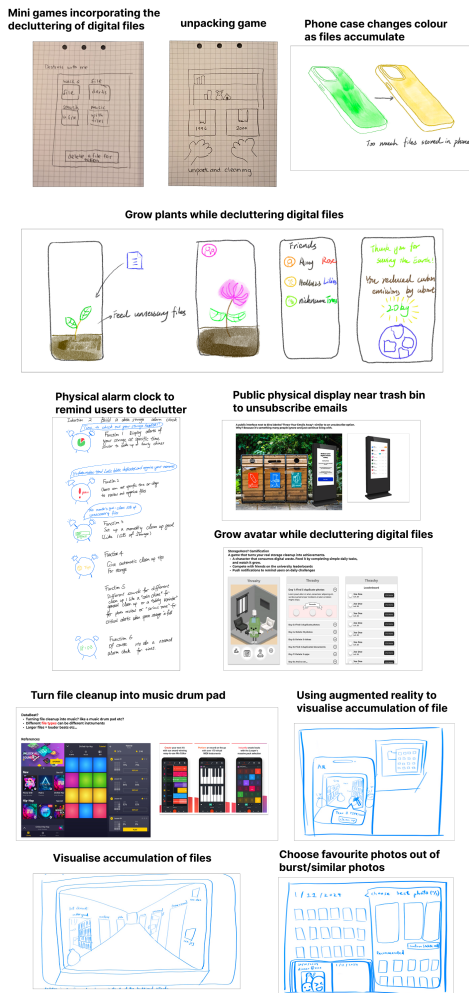


Figure 2: Crazy 8's showcasing rapid ideas focused on digital waste.

3.2 Interview

Based on the insights from the survey, semi-structured interviews were conducted with 5 participants, including both students and professors. Three themes emerged through the conduct of bottom-up thematic analysis.

The first theme is “learning about environmental impact as motivation”. Participants were unaware of the environmental impact of digital waste, but expressed interest in a solution that could provide related information. The theme echoed the survey result that participants lack environmental awareness and built on the findings by showing participants’ willingness to learn more about the impact of their habits.

The second theme is “excessive files complicate clean-up”. Many interviewees mentioned that they did not delete their files because they were overwhelmed by the volume or struggled to locate the

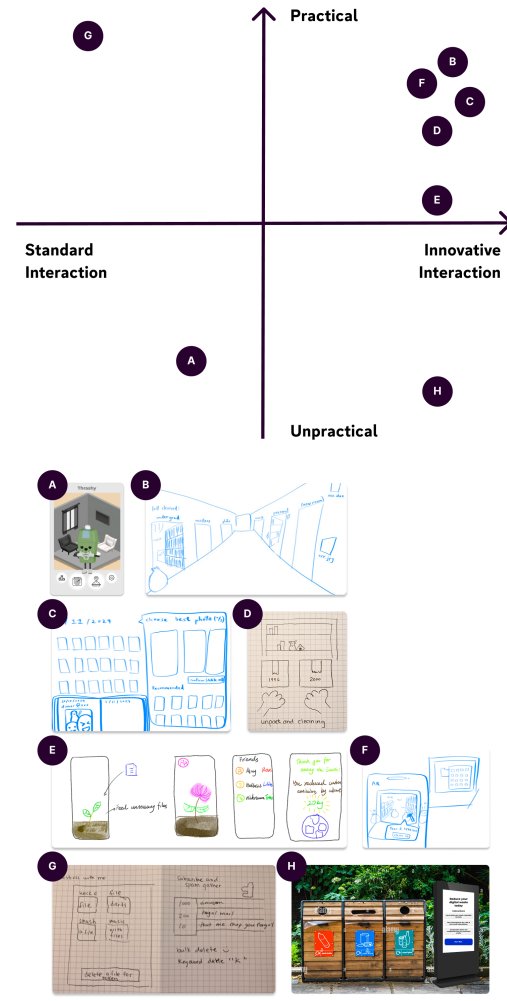


Figure 3: 2x2 matrix evaluating the crazy 8 ideations.

files they wanted to delete. This theme also aligned with the survey’s result of how users struggle to find files while providing more insight into the reason behind their frustration.

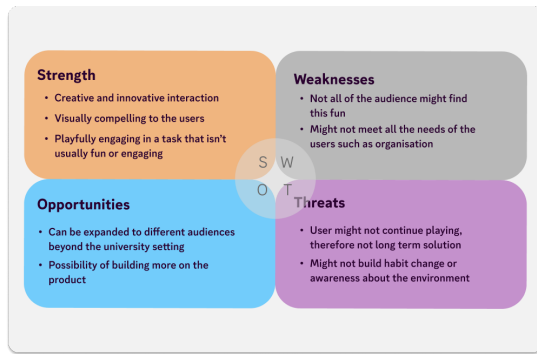
The last theme is “cleaning motivated by personal storage constraints”. Responses to motivation-related questions revealed that interviewees often declutter for personal needs, such as lack of storage, instead of environmental causes.

Personas (see Figure 1) were created based on the insights from the survey and interview to facilitate the design process and inform design decisions.

4 Design Process

4.1 Initial Ideation

With the target personas identified and the problem in mind, initial ideation began with ‘crazy 8s’ sketches (see Figure 2) in which a variety of ideas were proposed. Group voting and evaluation using 2x2 matrix (see Figure 3) based on creativity and practicality were then conducted to narrow down the ideas. Through group



(a) Gamified phone application.



(b) Email Extension.

Figure 4: SWOT analysis of gamified phone application (Up) and email extension (Down).

discussion, several ideas were combined, resulting in two potential designs.

The first design was a mobile application that used gamified interactions and visualisations to help users declutter old and unused files while highlighting the accumulation of digital files. The second design was a centralised email extension that assists users in easily unsubscribing from emails and deleting emails in bulk. Low-fidelity wireframes of both designs were created to better understand the possibility of how the two designs can facilitate digital decluttering and to better convey the two designs.

The two low-fidelity wireframes were evaluated using a Strength, Weakness, Opportunities, and Threats (SWOT) analysis (see Figure 4). The analysis showed that the two designs focused on different aspects of digital files and could neither sufficiently address awareness-building nor promote long-term habit change. It was then decided to combine the two designs to create a more comprehensive solution to tackle all types of files.

4.2 Second Ideation

After deciding to merge the concepts, more sketches were done to generate ideas for a design that could tackle all file types while promoting awareness and habit change. Through these additional sketches Digibyte was conceptualised, utilising elements from each that best fit together as a singular design solution. The main component extracted were: detecting duplicate files, displaying file data

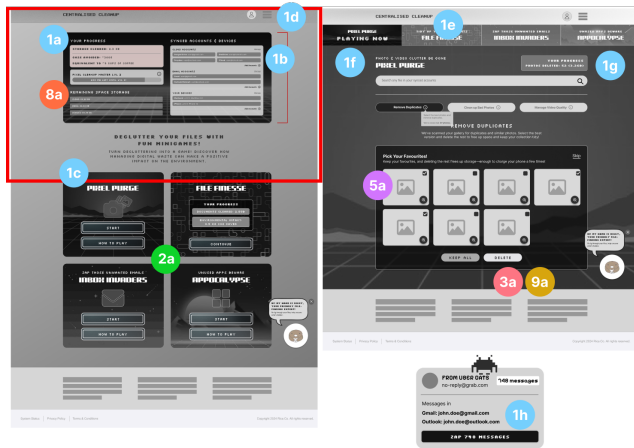
**Figure 5: The low-fidelity wireframe.**

such as date created, and a game-like design. A wireframe was developed to visualise the design concept after merging all the elements that had been selected. A web-based platform was used as all types of files can be accessed through computers.

The combined solution aimed to centralise physical and cloud storage, enabling users to declutter various digital files while learning about their environmental impact and promoting habit change for better digital waste management, aspects that were missing in the previous designs.

4.3 First Prototype

A prototype for Digibyte (see Figure 5) was developed to better visualise the design concept after merging the two ideas. A heuristic evaluation was then conducted to ensure all user features were intuitive and user-centred (see Figure 6). The analysis was conducted based on the Revised version (2014) of Nielsen's original heuristics,



1. Visibility of System Status

- The dashboard takes up too much space and users cannot see the main features when they first open the page.
- Synced accounts and devices section is too crowded and takes too much space.
- The name of each feature ("pixel purse", "file fitness", "inbox invaders" and "apocalypse") is confusing.
- The menu button does not have real use.
- Clicking the logo to return to the dashboard page is not intuitive for users.
- Tabs and sub-tabs are confusing.
- The progress section is easily overlooked.
- "Zap messages" is not intuitive, "delete" is recommended.

2. Visibility of System Status

- The background image of these four features is irrelevant to their function.

3. User control and freedom

- A pop up window to confirm deletion.

5. Error prevention

- Selected files (both photos and documents) should be highlighted

8. Aesthetic and minimalist design

- Dashboard should be simpler

9. Help users recognize, diagnose, and recover from errors

- A pop up window is useful for users to confirm to delete.

Figure 6: Results of the heuristic evaluation.

which revealed design flaws that were overlooked and provided guidance for improvements [11].

The heuristics analysis highlighted areas for improvement. Some of the important changes were implementing a pop-up for users to confirm if they wanted to delete the files they selected to prevent possible human error. The presentations of help documentation and instructions require changes to improve user-friendliness. The content on the "Your Progress" page was found to be still too overwhelming, and further adjustments to make the space more user-friendly were implemented.

After iterating the prototype according to the heuristic analysis, user testing was then conducted with 5 participants. The think-aloud method was used, where participants verbalised their thoughts while navigating the prototype and completing assigned tasks, such as "Delete 500 emails from your email inbox." Users

generally found the interface and interactions intuitive and straightforward, and many identified at least one feature they would use to declutter.

Participants appreciated the variety of features provided by the prototype. One participant highlighted that the email "unsubscribe" and bulk deletion would simplify their email decluttering. Other participant also expressed interest in utilising the platform's feature of uninstalling unused apps.

However, valuable insights revealed a gap in the design, several users expressed a desire for more information about their environmental impact. One user noted knowing the environmental background and comparison helps build their awareness, which demonstrates the importance to raise awareness in users.

The design was therefore refined to emphasise awareness and motivation. Pop-up notifications were introduced to celebrate users' decluttering progress, reinforcing positive behaviours. Additionally, a comprehensive information page about environmental impact was added to the dashboard, enabling users to track their progress and learn more about digital waste. These updates aim to both educate and inspire users to take meaningful action. Further testing was needed to evaluate their impact.

4.4 Second Prototype

After further iteration based on user insights, a second round of user testing was conducted with an additional five participants. Again, the think-aloud method was utilised.

The primary insights from this round of testing centred on usability. Users initially expressed confusion about device syncing, so a login page was added, and the home page was adjusted to clearly show connected devices. Users also requested more flexibility in files sorting, additional sorting options were therefore added. To prevent confusion, one set of navigation menus was. Lastly, an indication of storage space occupied by each file type was added, giving users a clearer indication of their progress.

4.5 System Usability Scale

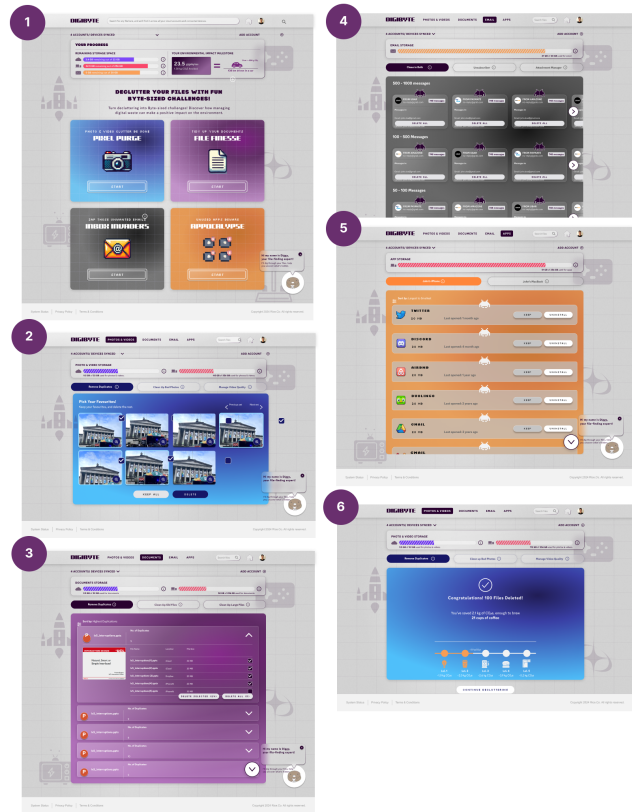
After conducting two rounds of user testing, we gradually refined our final design. To validate its effectiveness in addressing the identified issues, we invited the ten participants who had previously taken part in the user testing, along with an additional five individuals who had never used Digibyte before (a total of 15 participants), to experience the final design. Participants were asked to freely explore the system and complete a series of tasks, such as removing duplicate photos and unsubscribing from unnecessary email lists. Upon completing the experience, they were required to complete the System Usability Scale (SUS) [8], which consists of ten questions assessing the usability and learnability of a system. The final calculated SUS score for Digibyte was 78.83 (rounded to 2 d.p.) (see Figure 7), exceeding the benchmark of 68, indicating a high level of usability and learnability.

5 Final Design

After adopting the insights from the second round of user testing, a final prototype for Digibyte was developed. The final prototype features a landing page and an account-syncing page for first-time users. The landing page gives a brief overview of the website, while

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total
1. I think that I would like to use this system frequently.	4	5	3	4	4	3	4	5	3	4	5	4	2	4	5	
2. I found the system unnecessarily complex.	1	2	1	2	1	2	2	1	3	2	2	1	2	2		
3. I thought the system was easy to use.	5	3	5	4	3	4	5	4	4	5	4	4	4	5	4	
4. I think that I would need the support of a technical person to be able to use this system.	2	3	1	2	1	2	1	2	2	1	1	2	1	2	2	
5. I found the various functions in this system were well integrated.	5	4	5	4	4	4	4	4	3	4	5	4	4	5	4	
6. I thought there was too much inconsistency in this system.	1	1	1	2	1	1	3	2	1	1	1	2	1	2	1	
7. I would imagine that most people would learn to use this system very quickly.	5	3	5	4	3	4	4	5	3	3	3	4	4	4	4	
8. I found the system very cumbersome to use.	1	1	1	2	2	2	2	1	3	3	3	2	1	4	2	
9. I felt very confident using the system.	5	4	5	4	4	5	4	4	4	4	4	3	4	3	4	
10. I needed to learn a lot of things before I could get going with this system.	1	2	1	2	1	3	2	2	3	2	2	1	2	2	3	
Sum	95	75	95	75	80	72.5	80	85	62.5	77.5	80	75	80	72.5	78.83	

Figure 7: System Usability Scale Questions.

Figure 8: Digibyte Final Prototype Interfaces. (1) Homepage. (2-5) Category pages for photos, documents, emails and apps, respectively. (6) Milestone pages which displays saved CO₂e and its equivalent impact.

the syncing page allows first-time users to sync their storage and email accounts before starting the journey. The syncing of multiple storage spaces and accounts allows the centralisation of all files in one platform, which makes decluttering more convenient and efficient. The platform contains 5 main pages: Home, Photos and Videos, Documents, Email, and Apps.

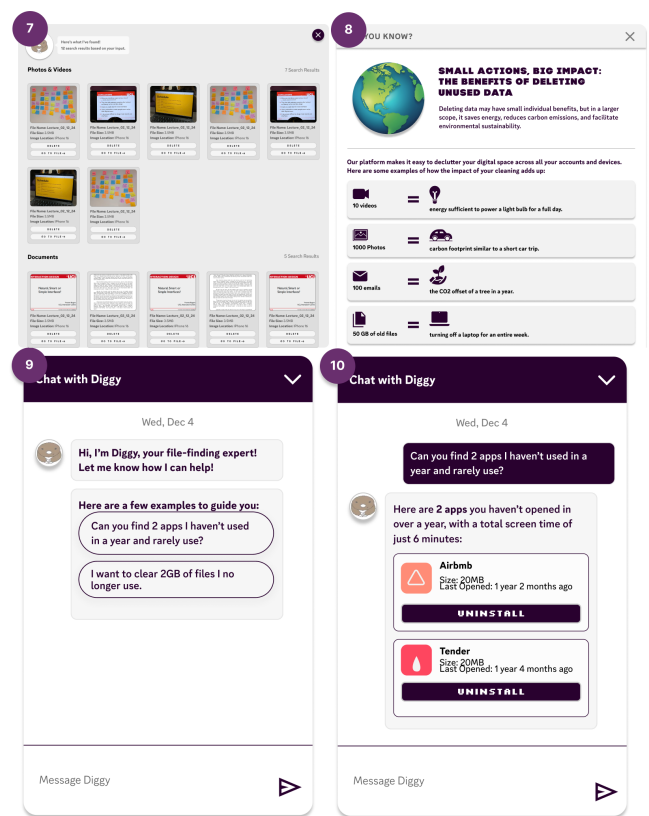


Figure 9: Digibyte Final Prototype Interfaces. (7) Smart Search (8) Awareness (9-10) Diggy the chatbot.

5.1 Home

The Home page (see Figure 8), with a progress dashboard, directs users to the different categories pages to begin users' journey of decluttering digital waste. Game-like names and elements were used to make the decluttering experience more engaging. The progress dashboard includes the users' decluttering progress in terms of both storage space and environmental impact. Progress bars indicate storage usage in the different synced accounts. From that, users receive a quick overview of the remaining storage in their different storage spaces. The dashboard also records users' progress of decluttering in terms of gigabytes, which is further translated into the carbon emission avoided and its implication in daily life. Equivalencing the vague number to real-life activities such as driving a car provides a better understanding of the tangible benefits of decluttering. If the users wish to learn more, additional information is provided in a pop-up window (see Figure 9).

5.2 Features

The main decluttering pages are categorised according to the types of digital files (see Figure 8). Every page contains a progress bar that indicates the storage space used by the corresponding types of digital files. As users delete more files, they unlock milestones that celebrate their efforts in decluttering digital waste (see Figure 8). As users reach a milestone, a message showing their progress

and its impact on the environment will be shown. By doing so, a sustainable habit of regular decluttering would hopefully be developed. The pages are designed to assist users in deleting digital waste by providing deletion suggestions without directly removing files. It helps initiate decluttering behaviour while retaining users' autonomy to decide what to delete. Each category page provides different deletion suggestions. As users declutter more, the storage they save ultimately benefits the environment.

5.3 Design Rationale

In photos, the website identifies similar or identical photos across storage spaces and suggests the users pick their favourites to keep and delete the rest. It also highlights blurry photos, old screenshots, and accidental shots. Furthermore, the platform identifies large videos across platforms and allows users to either compress, delete, or keep the video as is. Similar to photos, the website detects duplicates of documents across storage spaces and prompts users to delete them. Sorting options, such as the number of duplicates and file size, are provided to offer users flexible decluttering methods tailored to their preferences. The email tab simplifies email management with three key features. The "Bulk Delete" tab organises emails across email accounts by the sender and sorts them by the number of messages for bulk deletion. The "Unsubscriber" allows users to quickly unsubscribe from unwanted marketing and promotional emails. The "Attachment Manager" compiles all email attachments into one place, making it easy to review and delete unnecessary attachments. In applications, applications are synced from all devices and shown their occupied space and last opened date. Users can easily delete applications across devices.

The smart search function (see Figure 9) allows users to locate specific files across all storage spaces efficiently, tackling users' struggles in finding files, as revealed in user research. Across the website, users also have access to Diggy, the chatbot (see Figure 9), who helps automate the decluttering process through commands. Users can tell the chatbot they want to delete files, and the chatbot will compile them based on input.

Digibyte draws on Harold R. Hungerford's Behaviour Flow Chart, which emphasises the importance of "ownership" and "empowerment" in cultivating environmentally responsible behaviour [6]. The website builds on "ownership" by centralising users' physical and cloud storage, providing an overview of their files and supporting the identification and deletion of unnecessary items. To build a sense of "empowerment," Digibyte encourages users with celebratory messages after they delete digital waste. It also uses relatable comparisons, such as equating energy saved to cups of coffee, to highlight the user's achievement and develop an internal locus of control that their action can benefit the planet [5]. Digibyte strives to support users in developing into conscious consumers of digital files and help lower natural resource consumption from digital waste stored in clouds and physical devices.

6 Discussion and Next Steps

Digibyte promotes sustainable digital storage habits by centralising users' physical and cloud storage, making it easier to identify and delete unnecessary files such as photos, documents, emails, applications, and videos. The platform goes beyond being just a tool;

it aims to actively foster environmental awareness by providing environmental impact information and quantifying the amount of carbon emissions saved for day-to-day activities. Digibyte's long-term goal is to empower university affiliates to assess their storage needs, adopt sustainable file consumption practices, and integrate decluttering into their regular digital routine.

Limitations in this project may influence the long-term impact of its designs and there are future improvements to be made. First, the convenience sampling method used for user research and testing may have introduced bias, as participants were more likely to share similar demographics. Second, since our website requires users to connect their storage spaces and email accounts, concerns about data security and privacy may arise. Although the assurance of data security is not depicted in the current prototype, it should be noted that such concern was acknowledged throughout the design process. Future iterations will prioritise developing more robust features to enhance continuous awareness-building, encourage habit change, ensure data security and privacy, and expand accessibility. To foster long-term engagement, the focus will be on creating a model that sustains user motivation and the desire to create a positive impact on the environment. Potential approaches include introducing social interaction features to foster a sense of community to support each other in sustainable habits.

Digibyte is a tool that helps reduce the environmental impact of digital storage, especially in universities. By empowering institutional affiliates to declutter their files and make sustainable choices, the platform can help reduce the demand for additional cloud and physical storage, thereby minimising energy consumption and digital waste generation. Digibyte's potential can go beyond universities, as it can serve any individual with an excess of digital files and struggles to delete them in a user-friendly way.

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